

CH3421 CTP Assignment week 3

Héctor Maldonado - h.a.maldonadodeleon@tudelft.nl

Ramon van Valderen - r.d.vanvalderen@tudelft.nl

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1 Introduction

ANSYS Fluent[™] is a computer software program used for simulating and analyzing fluid dynamics, heat transfer, and other related phenomena. It is commonly used in engineering and scientific research to study the behavior of fluids (liquids and gases) in various applications and environments. With ANSYS Fluent[™] you can create a digital model of a physical system, define the relevant boundary conditions and properties, and then use mathematical equations to simulate the behavior of the fluids within that system. This allows you to study things like how heat is transferred, how fluid flows around objects, and how pressure changes affect the system.

Overall, ANSYS Fluent[™] is a powerful tool that helps engineers and researchers better understand how fluids behave in different situations, which can ultimately lead to the development of more efficient and effective systems and products.

In this session, we'll explore some of the capabilities of ANSYS Fluent[™] by simulating two steady-state systems involving laminar flows 1) in a pipe, and 2) passing over a cylinder.

2 Tutorial 1: Laminar flow in a pipe

2.1 Goals

- Create the geometry for the solution domain in SpaceClaim®
- Create the mesh for the solution domain using ANSYS Mesher®
- Develop the numerical solution to a laminar pipe flow problem in ANSYS Fluent
- Verify the numerical results from ANSYS Fluent

2.2 The problem

Consider fluid flowing through a circular pipe of constant radius as illustrated below (See Figure 1). The pipe diameter $D = 0.2$ m and length $L = 3$ m. Consider the inlet velocity to be constant over the cross-section and equal to 1 m/s. The pressure at the pipe outlet is 1 atm. Take density $\rho = 1$ kg/m³ and coefficient of viscosity $\mu = 2 \times 10^{-3}$ kg/m s. These parameters have been chosen to get a desired Reynolds number of 100 and don't correspond to any real fluid. Taken from: Bhaskaran.

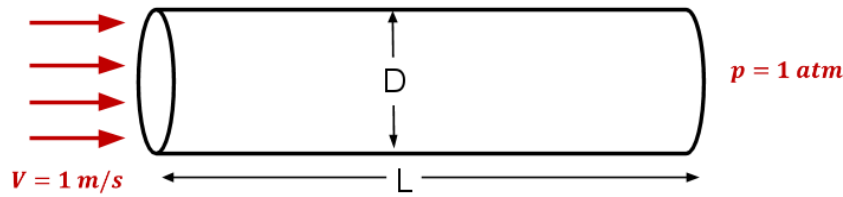


Figure 1: Sketch of the laminar flow in a pipe. **Note:** not to scale.

The aim of this exercise is to provide the following using ANSYS Fluent:

- Velocity magnitude contours
- Pressure contours
- Axial velocity profile at the outlet
- Axial velocity profile along the centerline of the pipe
- Skin drag coefficient at the pipe wall

2.3 Remarks and other tasks

In case you want to reproduce, generate or export the plots we obtained, you can take a look at the instructions provided in [Post-Processing - Tutorial: Laminar flow in a pipe](#). In all the cases, provide a suitable explanation of the profiles you obtained and how these relate (or not) to what you would expect or what has been reported in the literature.

3 Tutorial 2: Steady flow passing over a cylinder

3.1 Goals

- Create the geometry for the solution domain in SpaceClaim®
- Create the mesh for the solution domain using ANSYS Mesher®
- Develop the numerical solution to a laminar flow passing over a cylinder problem in ANSYS Fluent
- Verify the numerical results from ANSYS Fluent

3.2 The problem

Consider the steady-state case of a fluid flowing past a cylinder, as illustrated below (See Figure 2). Obtain the velocity and pressure distributions when the Reynolds number is chosen to be 20. In order to simplify the computation, the diameter of the cylinder is set to 1 m, the x component of the velocity is set to 1 m/s, and the density of the fluid is set to 1 kg/m³. Thus, the dynamic viscosity must be set to 0.05 kg/m s in order to obtain the desired Reynolds number. Taken from: Bhaskaran.

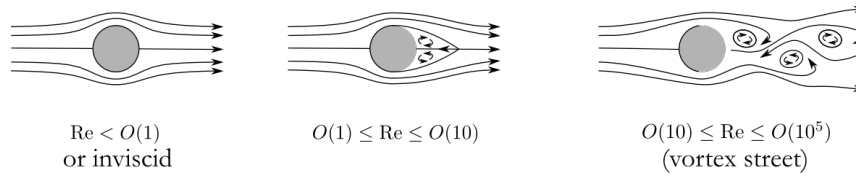


Figure 2: Sketch of the laminar flow passing over a cylinder at different flow regimes. **Note:** not to scale. Taken: Jaouen, [n.d.](#)

The aim of this exercise is to provide the following using ANSYS Fluent:

- Velocity magnitude contours
- Pressure contours
- Drag coefficient for the cylinder
- Pressure over the surface of the cylindrical surface (i.e., Pressure vs. θ)

3.3 Remarks and other tasks to reflect on

Please make sure you check the list below while following the tutorial:

- **Calculation of the drag coefficient:** The tutorial mentions that this can be done under **monitors**, however for the newer versions of Fluent this should be done via the **Report Definitions/Force reports**, there you will find **Drag**. Make sure you select **Drag coefficient**.
- **Reference values:** While elaborating on the drag coefficient you obtained from Fluent, look for and provide the equation Fluent solves for in order to calculate the drag coefficient and the relevance of the values provided under the *Reference Values* section for this calculation.
- **Validation of the results:** Compare the value estimated for the drag coefficient from your simulation with the ones available in the literature, and comment about any difference. Compare the plot you obtained in CFD-Post for the pressure vs. θ with the findings available in the literature (See: Figure 20 in J. Fluid Mech. (1980), vol. 98, part 4, pp. 819-855), and comment about the differences you might find.

4 Closing thoughts

Check the report outline available in BrightSpace to check the different sections and information you are asked to provide in each one of them. Make sure you provide all the required information and be creative in the way you convey your ideas and create a nice storyline out of the tutorials. Keep in mind that this report should contain two main sections corresponding to each one of the tutorials you solved. Also, make sure you provide explanations for the choices regarding the boundary conditions (e.g. magnitude and normal to boundary vs. per-components), numerical schemes (Second order upwind), models, monitors, etc. Be concise.

As a final advice, take a look at the book *Computational Fluid Dynamics for Engineers* from Andersson et al. to get to know more about the discretization schemes and their (dis-)advantages, some rules of thumb when setting up your models for different systems, and the best practices guidelines. Also, do not forget to take a look at the Users' and Theory Guides from ANSYS - you can find those in BrightSpace.

References

- Andersson, Bengt et al. (2012). *Computational Fluid Dynamics for Engineers*. Cambridge University Press. ISBN: 978-1-107-01895-2.
- Bhaskaran, Rajesh (2019). *FLUENT - Laminar Pipe Flow*. URL: <https://confluence.cornell.edu/display/SIMULATION/FLUENT++Laminar+Pipe+Flow>.
- (2022). *Steady Flow Past a Cylinder Using Ansys Workbench*. URL: <https://courses.ansys.com/index.php/courses/steady-flow-past-a-cylinder/>.
- Jaouen, Luc (n.d.). *Flow around a disk (or an infinite cylinder)*. URL: <https://apmr.matelys.com/BasicsMechanics/Fluid/FlowAroundADisk.html>.